

1 ^{1,†} and ^{1,†}

2 1
3 2

4 *corresponding author(s):

5 †these authors contributed equally to this work

6 ABSTRACT

7

8 Introduction

9 Model

10 Transitions between the presynaptic vesicle pools

11 Three pools of vesicles:

- 12 • Recycling pool
- 13 • Docked pool (i.e. readily releasable pool RRP)
- 14 • Fused pool

15 Calcium-dependent release probability (from docked to fused):

$$p_{rel} =$$

16 Calcium dynamics

$$\begin{aligned}\frac{dCa_{pre}}{dt} &= -\frac{Ca_{pre}}{\tau_{pre}} \\ \frac{dCa_{jump}}{dt} &= \frac{1 - Ca_{jump}}{\tau_{rec}} - \delta_{Ca} \cdot Ca_{jump} \cdot Ca_{pre} \\ Ca_{pre} &\leftarrow Ca_{pre} + Ca_{jump} \text{ upon spike arrival}\end{aligned}$$

Table 1. Model variables.

Symbol	Description	Unit
R	Number of vesicles in the recycling pool	-
D	Number of docked vesicles	-
F	Number of fused vesicles	-
Ca_{pre}	Intracellular presynaptic calcium concentration	mM
Ca_{jump}	Jump in Ca_{pre} upon spike arrival	mM

Table 2. Model parameters.

Symbol	Description	Unit	Experimental litterature
R_0	Recycling pool capacity	-	5-20 vesicles per bouton in cultured rat hip neurons ^{1,2}
D_0	Docked pool capacity	-	
$\tau_{R \rightarrow D}$	Vesicle transition rate from recycling to docked	s	
$\tau_{D \rightarrow R}$	Vesicle transition rate from docked to recycling (mixing)	s	
$\tau_{F \rightarrow R}$	Vesicle transition rate from fused to recycling	s	
τ_{pre}	Ca_{pre} attenuation recovery rate	s	
τ_{rec}	Calcium jump recovery rate	s	
δ_{Ca}	Jump attenuation fraction	$s^{-1} mM^{-1}$	
Sigmoid parameters			

References

1. Rizzoli, S. O. & Betz, W. J. Synaptic vesicle pools. *Nat. Rev. Neurosci.* **6**, 57–69, [10.1038/nrn1583](https://doi.org/10.1038/nrn1583) (2005).
2. Schikorski, T. & Stevens, C. F. Morphological correlates of functionally defined synaptic vesicle populations. *Nat. Neurosci.* **4**, 391–395, [10.1038/86042](https://doi.org/10.1038/86042) (2001).